

University of Minnesota
School of Physics and Astronomy

2026 Fall Mathematics 8302
Differential Topology
Assignment Solution

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Question 1.7

Let N denote the **north pole** $(0, \dots, 0, 1) \in \mathbb{S}^n \subseteq \mathbb{R}^{n+1}$, and let S denote the **south pole** $(0, \dots, 0, -1)$. Define the **stereographic projection** $\sigma : \mathbb{S}^n \setminus \{N\} \rightarrow \mathbb{R}^n$ by

$$\sigma(x_1, \dots, x_{n+1}) = \frac{(x_1, \dots, x_n)}{1 - x_{n+1}}. \quad (1)$$

Let $\tilde{\sigma} = -\sigma(-x)$ for $x \in \mathbb{S}^n \setminus \{S\}$.

- (a) For any $x \in \mathbb{S}^n \setminus \{N\}$, show that $\sigma(x) = u$, where $(u, 0)$ is the point where the line through N and x intersects the linear subspace where $x_{n+1} = 0$. Similarly, show that $\tilde{\sigma}$ is the point where the line through S and x intersects the same subspace. (For this reason, $\tilde{\sigma}$ is also called **stereographic projection from the south pole**.)
- (b) Show that σ is bijective and

$$\sigma^{-1}(u^1, \dots, u^n) = \frac{(2u^1, \dots, 2u^n, |u|^2 - 1)}{|u|^2 + 1}. \quad (2)$$

- (c) Compute the transition map $\tilde{\sigma} \circ \sigma^{-1}$ and verify that the atlas consisting of the two charts $(\mathbb{S}^n \setminus \{N\}, \sigma)$ and $(\mathbb{S}^n \setminus \{S\}, \tilde{\sigma})$ defines a smooth structure on $\mathbb{S}^n$. (The coordinates defined by σ and $\tilde{\sigma}$ are called **stereographic coordinates**.)
- (d) Show that this smooth structure is the same as the one defined in Example 1.31.

Answer

Question 1.9

Answer